

Describe how data validation and data verification help protect the integrity of data

9618_w23_qp_13_sc_8.a

QUESTION

- 8 (a) Data verification is one method of protecting the integrity of data.

Describe **one** other method of protecting the integrity of data.

.....

.....

.....

..... [2]

MARK SCHEME

8(a)	1 mark for each bullet point (max 2) • Validation // a validation method named or described • ...protects the data by ensuring that the data is reasonable / sensible and within specified bounds	2
------	---	----------

9618_s23_qp_13_sc_4.d.iii

QUESTION

- (iii) State why the car registration number might be incorrect even after it has been validated and verified.

.....

..... [1]

MARK SCHEME

4(d)(iii)	The registration number on the original document might be in the correct format but may be the incorrect registration number for that car.	1
-----------	--	----------

QUESTION

- 4 (a) State the difference between **data verification** and **data validation**.

.....

 [1]

MARK SCHEME

4(a)	data verification is checking if input data is the same as the original whereas data validation is checking that the data is reasonable / sensible	1
------	--	----------

QUESTION

- (b) Xanthe uses both data validation and data verification when entering data on her computer.

- (i) Describe how data validation helps to protect the integrity of the data. Give an example in your answer.

Description

Example [2]

MARK SCHEME

2(b)(i)	1 mark per bullet point - validation checks that data is reasonable / sensible - example e.g. checking data is the right number / type of characters	2
---------	--	----------

QUESTION

- (ii) Describe how data verification helps to protect the integrity of the data. Give an example in your answer.

Description

.....

.....

Example

[2]

MARK SCHEME

2(b)(ii)	1 mark per bullet point • verification checks that data is the same as the original • by example e.g. double entry	2
----------	--	----------

Describe and use methods of data validation

9618_w25_qp_13_sc_7.e

QUESTION

- (e) The table contains three algorithms that perform data validation. The third algorithm uses the functions `MID` and `LENGTH`.

The functions `MID` and `LENGTH` are defined as follows:

- `MID(ThisString : STRING, x : INTEGER, y : INTEGER)` RETURNS STRING
returns a string of length `y` starting at position `x` from `ThisString`
- `LENGTH(ThisString : STRING)` RETURNS INTEGER
returns the integer value representing the length of `ThisString`

Complete the table to identify the method of data validation for each algorithm. Each method of data validation must be different.

Algorithm	Method of data validation
<pre>INPUT x IF x < 1 or x > 26 THEN OUTPUT "Invalid" ENDIF</pre>	
<pre>INPUT x IF x <> 'R' AND x <> 'G' AND x <> 'B' THEN OUTPUT "Invalid" ENDIF</pre>	
<pre>INPUT x FLAG ← FALSE FOR INDEX ← 1 TO LENGTH(x) IF MID(x, INDEX, 1) = "@" THEN FLAG ← TRUE ENDIF NEXT INDEX IF NOT FLAG THEN OUTPUT "Invalid" ENDIF</pre>	

[3]

MARK SCHEME

7(e)	1 mark for each validation check correctly identified, max 3 marks Range Check Existence Check Format Check	3
------	--	----------

9618_w23_qp_13_sc_8.a

QUESTION

8 (a) Data verification is one method of protecting the integrity of data.

Describe **one** other method of protecting the integrity of data.

.....

.....

.....

..... [2]

MARK SCHEME

8(a)	1 mark for each bullet point (max 2) • Validation // a validation method named or described • ...protects the data by ensuring that the data is reasonable / sensible and within specified bounds	2
------	---	----------

QUESTION

(d) The data is validated and verified when it is entered into the database.

- (i) The car registration number must be: 1 letter, followed by 3 numbers, followed by 2 letters.

For example, A123AA is valid but A12AA is invalid.

One way that a registration number can be validated is by using a presence check to make sure the registration number has been entered.

Describe **two other** ways that the car registration number can be validated.

1

.....

2

.....

[2]

MARK SCHEME

4(d)(i)	1 mark each to max 2 - Length check: the registration number must be 6 characters long - Format check: the registration number must be in the format letter-digit-digit-digit-letter-letter - Type check: the registration number must be alphanumeric	2
---------	--	----------

QUESTION

(c) The database `Lessons` has the following tables:

`HORSE(HorseID, Name, Height, Age, HorseLevel)`

`STUDENT(StudentID, FirstName, LastName, RiderLevel, PreferredHorseID)`

`LESSON(LessonID, Date, Time, StudentID, HorseID, LessonContent)`

Dates in this database are stored in the format `#DD/MM/YYYY#`.

The fields `RiderLevel` and `HorseLevel` can only have the values: `Beginner`, `Intermediate` or `Advanced`.

(i) Describe **two** methods of validating the field `RiderLevel`.

- 1
-
-
- 2
-
-

[2]

MARK SCHEME

2(c)(i)	1 mark each to max 2 • Presence check to make sure that the (rider level) is entered • Look-up / Existence check to make sure the rider level is only <code>Beginner</code>, <code>Intermediate</code> or <code>Advanced</code> • Length check to make sure the rider level entered is either 8 or 12 characters • Type check to make sure the rider level is alphanumeric	2
---------	--	----------

QUESTION

(c) One validation method is a presence check.

Describe **two other** validation methods that can be used to validate non-numeric data.

1

.....

2

.....

[2]

MARK SCHEME

4(c)	1 mark for each bullet point (max 2). For example: - to make sure data is in the required format // only expected characters allowed - to make sure the data is already present in the system - to make sure the data contains the correct number of characters - to ensure that non-numeric data is entered	2
------	--	----------

QUESTION

(b) Xanthe uses both data validation and data verification when entering data on her computer.

(i) Describe how data validation helps to protect the integrity of the data. Give an example in your answer.

Description

.....

.....

Example

[2]

MARK SCHEME

2(b)(i)	1 mark per bullet point - validation checks that data is reasonable / sensible - example e.g. checking data is the right number / type of characters	2
---------	--	----------

9618_s21_qp_11_sc_6.a

QUESTION

6 Each of the following algorithms performs data validation.

State the type of validation check that each of the algorithms performs.

(a)

```
INPUT x
```

```
IF x < 0 OR x > 10 THEN
```

```
    OUTPUT "Invalid"
```

```
ENDIF
```

..... [1]

MARK SCHEME

6(a)	Range (check)	1
------	---------------	---

9618_s21_qp_11_sc_6.b

QUESTION

(b)

```
INPUT x
```

```
IF x = "" THEN
```

```
    OUTPUT "Invalid"
```

```
ENDIF
```

..... [1]

MARK SCHEME

6(b)	Presence (check)	1
------	------------------	---

QUESTION

(c)

```
INPUT x
```

```
IF NOT(x = "Red" OR x = "Yellow" OR x = "Blue") THEN
```

```
    OUTPUT "Invalid"
```

```
ENDIF
```

..... [1]

MARK SCHEME

6(c)	Existence (check)	1
------	-------------------	----------

Describe and use methods of data verification during data entry and data transfer

9618_w25_qp_13_sc_7.c

QUESTION

(c) Identify **and** describe **one** method of data verification that can be used when transferring data from the digital video camera to the server.

Method

Description

.....

.....

.....

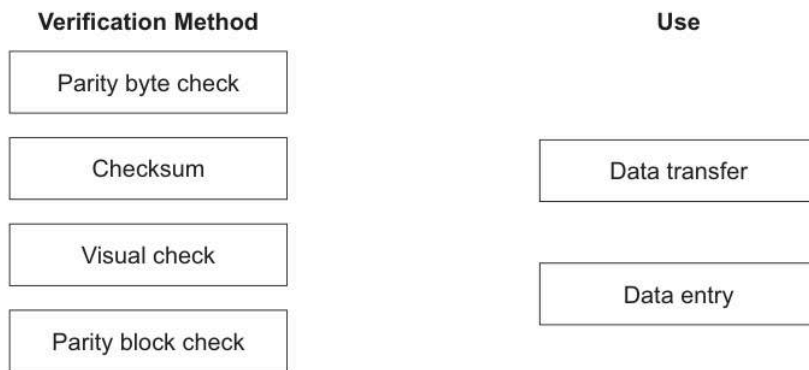
[3]

MARK SCHEME

7(c)	1 mark for the name 2 marks for the matching description, max 3 marks - Parity Byte Check - a parity bit is added to each byte to make the number of 1s match the parity, odd or even - each byte can be checked on receipt and request to be resent if the byte does not match parity - Parity Block Check - a bit is added to each byte, but a parity byte is also set for each block - the location of an error can be found using vertical and horizontal parity - Checksum - A calculation is made from the data and transmitted with the data - The receiver performs the same calculation and compares with received checksum to see if they match	3
------	---	---

QUESTION

- 1 Draw **one** line from each verification method to indicate whether it is used during data transfer or data entry.



[2]

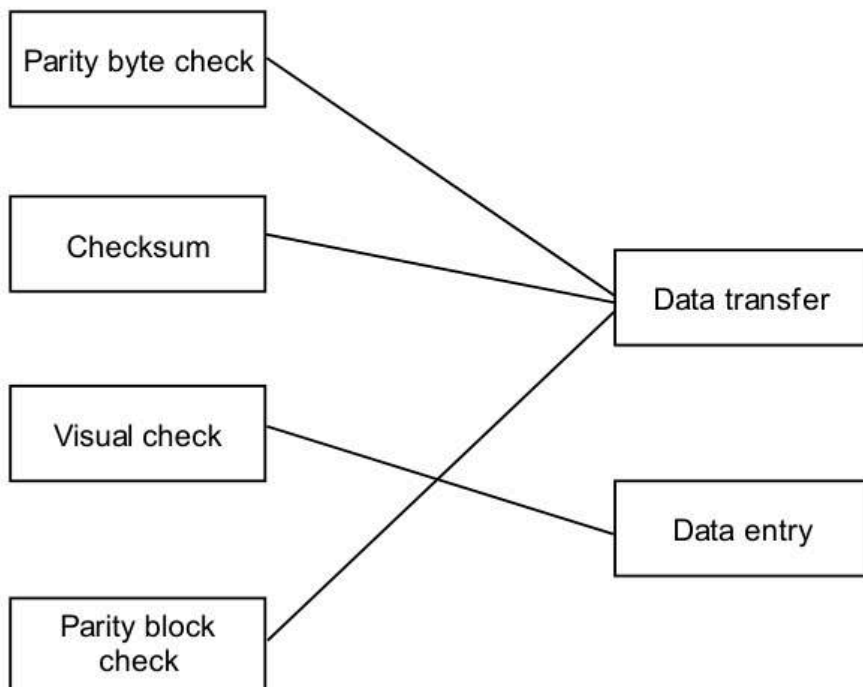
MARK SCHEME

1

1 mark for 2 or 3 correctly connected verification method boxes, **2 marks** for all 4 correct

Verification Method

Use



2

QUESTION

(c) A computer system uses even parity. The least significant (rightmost) bit of each byte is the parity bit.

(i) Complete the byte by writing the missing parity bit:

								parity bit ↓
0	1	0	1	1	1	0		

[1]

MARK SCHEME

6(c)(i)	1 mark for correct answer	1								
parity bit ↓ <table><tr><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td></tr></table>			0	1	0	1	1	1	0	0
0	1	0	1	1	1	0	0			

QUESTION

(ii) The computer also uses parity block check. The parity block check uses even parity. Computer A transmits four bytes of data to computer B, followed by a parity byte. Computer B receives the following sequence of bytes.

								parity bit ↓
	1	0	1	1	0	1	1	1
	0	1	1	1	0	0	0	0
	0	0	0	1	1	0	1	1
	0	1	1	1	0	1	0	0
parity byte →	1	0	1	0	0	0	0	0

Following transmission, one of the four bytes of data has an error in one of the bits.

Circle the bit that has been altered during the data transfer.

[1]

MARK SCHEME

6(c)(ii)

1 mark for correct answer

parity bit
↓

	1	0	1	1	0	1	1	1
	0	1	1	1	0	0	0	0
	0	0	0	1	1	0	1	1
	0	1	1	1	0	1	0	0
parity byte→	1	0	1	0	0	0	0	0

1

[1]

9618_s25_qp_11_sc_7.c

QUESTION

(c) The data that is transferred can also be verified using a checksum.

Explain how data can be verified using a checksum.

.....

.....

.....

.....

.....

..... [3]

MARK SCHEME

7(c)	1 mark each to max 3 • The data is put through an algorithm to create a checksum value • The data and checksum are sent to the receiver • The receiver performs the same algorithm on the data • if both checksums match the data is verified	3
------	--	---

QUESTION

16

- (f) The data from the robots is transmitted to a central computer using a wireless connection.
- (i) Complete the table by identifying **and** describing **two** methods of data verification that can be used during data transfer.

	Method	Description
1		
2		

[4]

MARK SCHEME

7(f)(i)	1 mark for each correct method and 1 mark for corresponding description to max 4: <table><tr><th>Method</th><th>Description</th></tr><tr><td>Parity byte</td><td>An additional bit is added to make the number of 1s in the byte odd or even to match the parity. If a byte with an odd number of 1 bits is received when even parity is used, there is an error.</td></tr><tr><td>Parity block</td><td>Parity is calculated horizontally and vertically. A parity byte is created from the bits produced by the vertical parity check. This is sent with the data. The parity is re-checked when received and the position of an incorrect bit can be determined.</td></tr><tr><td>Checksum</td><td>A calculation is made from the data and the result transmitted with the data. The receiver repeats the calculation and compares the result with the value received. If the two are different, there is an error.</td></tr></table>	Method	Description	Parity byte	An additional bit is added to make the number of 1s in the byte odd or even to match the parity. If a byte with an odd number of 1 bits is received when even parity is used, there is an error.	Parity block	Parity is calculated horizontally and vertically. A parity byte is created from the bits produced by the vertical parity check. This is sent with the data. The parity is re-checked when received and the position of an incorrect bit can be determined.	Checksum	A calculation is made from the data and the result transmitted with the data. The receiver repeats the calculation and compares the result with the value received. If the two are different, there is an error.	4
Method	Description									
Parity byte	An additional bit is added to make the number of 1s in the byte odd or even to match the parity. If a byte with an odd number of 1 bits is received when even parity is used, there is an error.									
Parity block	Parity is calculated horizontally and vertically. A parity byte is created from the bits produced by the vertical parity check. This is sent with the data. The parity is re-checked when received and the position of an incorrect bit can be determined.									
Checksum	A calculation is made from the data and the result transmitted with the data. The receiver repeats the calculation and compares the result with the value received. If the two are different, there is an error.									

9618_s24_qp_11_sc_5.b

QUESTION

- (b) The bank wants to protect the integrity of its data while transferring the data to other banks. Parity check is one example of data verification.

Complete the description of parity check when Computer A is transmitting data to Computer B.

Computer A and Computer B agree on whether to use parity. Computer A divides the data into groups of The number of 1s in each group is counted. If the agreed parity is and the group has an even number of 1s, a parity bit of 1 is appended, otherwise a parity bit of 0 is appended.

In a parity check the bytes are grouped together, for example in a grid. The number of 1s in each column (bit position) is counted. A bit is assigned to each column to make the column match the parity. These parity bits are transmitted with the data as a parity

[5]

MARK SCHEME

5(b)	1 mark for each correctly completed term: • odd or even • 7-bits • odd • block • byte Computer A and Computer B agree on whether to use odd or even parity. Computer A divides the data into groups of 7-bits. The number of 1s in each group is counted. If the agreed parity is odd and the group has an even number of 1s a parity bit of 1 is appended, otherwise a parity bit of 0 is appended. In a parity block check the bytes are grouped together, for example in a grid. The number of 1s in each column (bit position) is counted. A bit is assigned to each column to make the column match the parity. These parity bits are transmitted with the data as a parity byte.	5
------	---	----------

9618_w23_qp_13_sc_3.c

QUESTION

(c) An example of a tutor ID is NK16C6.

An administrative officer enters the tutor ID into the `TUTOR` table.

Explain how data verification can be used when the tutor ID is entered.

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

MARK SCHEME

3(c)	1 mark for each bullet point. • The administrator completes a visual check / checks by eye • ...that the tutor identifier input matches the tutor identifier on the original document • Double entry check // the administrator (or a second person) enters the number a second time • ...and the system compares it with the first entry	4
------	--	----------

9618_s23_qp_13_sc_4.d.ii

QUESTION

(ii) Describe **two** ways that the car registration number can be verified when it is entered into the database.

1

.....

2

.....

[2]

MARK SCHEME

4(d)(ii)	1 mark each • Visual check: Manually compare the registration number entered with the source document • Double entry: Enter the registration number twice and the computer compares to check they are the same	2
----------	---	----------

QUESTION

- (b) A checksum can be used to detect errors during data transmission.

Describe how a checksum is used.

.....

.....

.....

.....

.....

..... [3]

MARK SCHEME

4(b)	1 mark for each bullet point (max 3): - checksum value is calculated from the data before transmission // correct description of a checksum algorithm ... this calculated value is transmitted with the data receiving computer recalculates the checksum from the received data - if the checksum received and calculated match, no error has occurred // if the checksum received and calculated do not match, an error has occurred	3
------	---	----------

QUESTION

- (ii) Describe how data verification helps to protect the integrity of the data. Give an example in your answer.

Description

.....

.....

Example

[2]

MARK SCHEME

2(b)(ii)	1 mark per bullet point - verification checks that data is the same as the original - by example e.g. double entry	2
----------	--	----------